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PERFORMANCE SPECIFICATION

PUMPS AND MOTORS, HYDRAULIC
(NAVAL SHIPBOARD USE)

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers hydraulic pumps and motors used as components of Naval shipboard machinery, such as steering gear, cranes, elevators, winches, windlasses, and hoists, that rely upon these pumps and motors for delivery or consumption of power using hydraulic fluid or oil.

1.2 Classification. Pumps and motors are of the following classes, as specified (see 6.2):

a. Class I – Axial piston, fixed or variable delivery
b. Class II – Radial piston, fixed or variable delivery
c. Class III – Vane, fixed delivery

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-901	-	Shock Tests, H.I. (High-Impact) Shipboard Machinery, Equipment, and Systems, Requirements for
MIL-PRF-6083	-	Hydraulic Fluid, Petroleum Base, for Preservation and Operation

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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MIL-DTL-15024	-	Plates and Tags for Identification of Equipment, General Specification for
MIL-PRF-17331	-	Lubricating Oil, Steam Turbine and Gear, Moderate Service
MIL-PRF-17672	-	Hydraulic Fluid, Petroleum, Inhibited
MIL-DTL-24441	-	Paint, Epoxy-Polyamide, General Specification for
MIL-DTL-24441/20	-	Paint, Epoxy-Polyamide, Green Primer, Formula 150, Type III
MIL-DTL-24441/21	-	Paint, Epoxy-Polyamide, Haze Gray Formula 151, Type III
MIL-C-24707/5	-	Castings, Ductile Iron and Austenitic Ductile Iron
MIL-DTL-32353	-	Hydraulic & Lubricating Oil, Synthetic Hydrocarbon Base

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-167-1	-	Mechanical Vibrations of Shipboard Equipment (Type I – Environmental and Type II – Internally Excited)
MIL-STD-740-2	-	Structureborne Vibratory Acceleration Measurements and Acceptance Criteria of Shipboard Equipment
MIL-STD-810	-	Environmental Engineering Considerations and Laboratory Tests
MIL-STD-1474	-	Noise Limits
MIL-STD-2193	-	Ship Hydraulic System Components

(Copies of these documents are available online at https://quicksearch.dla.mil/.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

NAVAL SEA SYSTEMS COMMAND (NAVSEA) PUBLICATIONS

T9070-AL-DPC-020/077-2	-	NAVSEA Hazardous Material Avoidance Process

(Copies of this document are available online via Technical Data Management Information System (TDMIS) at https://mercury.tdmis.navy.mil/ by searching for the document number without the suffix. Refer questions, inquiries, or problems to: DSN 296-0669, Commercial (805) 228-0669. This document is available for ordering (hard copy) via the Naval Logistics Library at https://nll.navsup.navy.mil. For questions regarding the NLL, contact the NLL Customer Service at nllhelpdesk@navy.mil, (866) 817-3130, or (215) 697-2626/DSN 442-2626.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM D6304	-	Standard Test Method for Determination of Water in Petroleum Products, Lubricating Oils, and Additives by Coulometric Karl Fischer Titration

(Copies of this document are available online at www.astm.org.)

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO 281	-	Rolling bearings – Dynamic load ratings and rating life

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ISO 3019-1 - Hydraulic fluid power – Dimensions and identification code for mounting flanges and shaft ends of displacement pumps and motors – Part 1: Inch series shown in metric units

ISO 3019-2 - Hydraulic fluid power – Dimensions and identification code for mounting flanges and shaft ends of displacement pumps and motors – Part 2: Metric series

ISO 4406 - Hydraulic fluid power – Fluids - Method for coding the level of contamination by solid articles

ISO 11171 - Hydraulic fluid power – Calibration of automatic particle counters for liquids

(Copies of these documents are available online at www.iso.org.)

SAE INTERNATIONAL

SAE J744 - Hydraulic Pump and Motor Mounting and Drive Dimensions

SAE J2321/2 - Disposable Hydraulic Filter Element – 190 L/min Flow, 21 MPa Collapse Pressure, Filtration Ratio – 75 at 7µm

(Copies of these documents are available online at www.sae.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article inspection. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with table I and 4.2.

3.2 Materials. Materials shall be selected that are capable of meeting all operational and environmental requirements specified herein. Materials shall be compatible with specified fluids and temperatures. The hydraulic pumps and motors shall not contain any radioactive materials or chemicals categorized as “prohibited” in accordance with T9070-AL-DPC-020/077-2. The materials specified in MIL-STD-2193, to include metals, plated coatings, plastic parts, seals, and fasteners, are recommended but are not mandatory.

3.2.1 Metals. Metals that do not inherently possess corrosion-resisting characteristics shall be protected to resist corrosion that may result from conditions such as dissimilar metal combinations, moisture, salt spray, or high-temperature deterioration (see 3.3, 3.5.7, 3.8.1, and 6.4.1). Metals shall be compatible with the intended temperature, function, service, and storage conditions to which the pumps and motors are exposed.

3.2.1.1 Prohibited metals. The following metals shall not be used:

a. Cadmium-plated and zinc-plated parts in contact with hydraulic fluid.

b. Cadmium or zinc materials.

c. Magnesium or magnesium-based alloys.

d. Beryllium or metal alloys containing greater than 3 percent beryllium.

e. Mercury or mercury-bearing parts.

f. Aluminum alloys that are not resistant to stress corrosion cracking.

g. Low ductility castings, which have less than 18 percent elongation in a standard tension test, used in pressure-containing boundaries of hydraulic pumps or hydraulic motors. Castings specified in MIL-C-24707/5 are recommended but are not mandatory.

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3.2.2 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.3 Environmental conditions.

3.3.1 Salt spray or fog. Pumps and motors shall withstand exposure to a salt-laden environment for 48 hours without incurring corrosive damage or degradation in operational performance below specified limits. Compliance with the above requirements on similar equipment with the same base and coating materials and equivalent coating thickness may be used to satisfy this requirement.

3.3.2 Vibration. Pumps and motors shall withstand external shipboard environmental vibration in accordance with MIL-STD-167-1, type I, without damage to parts, loosening of parts, or degradation of operational performance (including leakage) below specification requirements. Pumps and motors shall be balanced to minimize internally excited vibration, bearing wear, and noise in accordance with MIL-STD-167-1, type II. Both type I and type II criteria shall be met for speeds up to and including 115 percent of rated speed (see 4.6.7.2).

3.3.3 High-impact shock. Pumps and motors shall meet the requirements for the specified shock grade and class (see 6.2), and shall be verified as specified in 4.6.7.3. If a shock test is required, a momentary interruption of the pump or motor function or leakage during shock loading shall not be interpreted as a failure.

3.4 Performance characteristics.

3.4.1 Operating conditions. Unless otherwise specified (see 6.2), pumps and motors shall operate under the following pressure, speed, and temperature conditions.

3.4.1.1 Pressures.

3.4.1.1.1 Designed working pressure (see 6.5.1). Pumps and motors shall operate with continuous internal pressures up to and including 3,000 pounds per square inch gauge (PSIG) for classes I and II and 2,000 PSIG for class III at discharge for pumps and at suction for motors without evidence of damage or impaired performance (see 3.4.2 and 3.4.3).

3.4.1.1.2 Maximum design pressure (see 6.5.2). Pumps and motors shall operate with intermittent internal pressures of up to and including 4,500 PSIG for classes I and II and 3,000 PSIG for class III at discharge for pumps and at suction for motors without evidence of damage or impaired performance (see 3.4.2 and 3.4.3).

3.4.1.2 Speeds.

3.4.1.2.1 Designed working or rated speed. Hydraulic pumps and motors shall operate at the continuous working speed as specified (see 6.2) without evidence of damage or impaired performance (see 3.4.2 and 3.4.3).

3.4.1.2.2 Minimum steady speed (motors). Hydraulic motors shall operate at the following continuous speeds against a maximum of 20 percent of the rated load without cogging or indication of ±5 percent speed variation from the applicable steady speed listed below:

a. Class I motors: 150 revolutions per minute (rpm)

b. Class II motors: 10 rpm

c. Class III motors: 75 rpm

3.4.1.3 Temperatures. Hydraulic pumps and motors shall operate with a suction or inlet temperature within the range of 90 to 160 °F without evidence of damage or impaired performance characteristics (see 3.4.2 and 3.4.3).

3.4.2 Pumping/motoring characteristics.

3.4.2.1 Pumping characteristics (pumps). As specified (see 6.2), pumps shall provide flow rate output for the operating conditions specified in 3.4.1.

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3.4.2.2 Motoring characteristics (motors). As specified (see 6.2), motors shall provide torque output for the operating conditions specified in 3.4.1.

3.4.3 Efficiency. Unless otherwise specified (see 6.2), pumps and motors shall exhibit the following efficiencies.

3.4.3.1 Mechanical efficiency. Pumps and motors shall exhibit a mechanical efficiency (see 6.5.5 for pumps and 6.5.3 for motors) of not less than 85 percent under the operating speed, suction pressure, and normal oil temperature conditions as specified in paragraphs 4.5.1, 4.5.2, and 4.5.3.1, respectively, at the designed working pressure (see 6.5.1). Pumps and motors shall exhibit a mechanical efficiency of not less than 80 percent under the operating speed, suction pressure, and hot oil temperature conditions as specified in 4.5.1, 4.5.2, and 4.5.3.2, respectively, at the designed working pressure (see 6.5.1).

3.4.3.2 Volumetric efficiency. Pumps and motors shall exhibit a volumetric efficiency (see 6.5.6 for pumps and 6.5.4 for motors) of not less than 90 percent under the operating speed, suction pressure, and normal oil temperature conditions as specified in paragraphs 4.5.1, 4.5.2, and 4.5.3.1, respectively, at the designed working pressure (see 6.5.1). Pumps and motors shall exhibit a volumetric efficiency of not less than 85 percent under the operating speed, suction pressure, and hot oil operating conditions as specified in 4.5.1, 4.5.2, and 4.5.3.2, respectively, at the designed working pressure (see 6.5.1).

3.5 Design/construction.

3.5.1 Endurance. Pumps and motors shall not be damaged or have their performance characteristics (see 3.4.2 and 3.4.3) impaired by 100 hours of load service, as specified in either 4.6.4.1 or 4.6.4.2, as applicable.

3.5.2 Proof pressure. Unless otherwise specified (see 6.2), pumps and motors shall withstand the following proof pressures.

3.5.2.1 Proof pressure main flow ports. Pumps and motors shall withstand a static internal pressure of 6,000 PSIG for classes I and II and 4,000 PSIG for class III at discharge ports for pumps and at suction ports for motors without evidence of damage, permanent deformation, or leakage. For design guidance on material strength factor of safety, see 6.4.2.

3.5.2.2 Proof pressure case drain ports. Pumps and motors shall withstand a static internal pressure of 60 PSIG at case drain ports without evidence of damage, permanent deformation, or leakage. This requirement shall not apply to class III auxiliary pumps.

3.5.3 Joint tightness (leakage). Pumps and motors shall operate at the designed working pressure and rated speed without external leakage at static seal joints and shall not exceed leakage of 1 drop per 5 minutes at shaft seals.

3.5.4 Direction of rotation. Unless otherwise specified (see 6.2), pumps shall operate in the clockwise direction of rotation. Motors shall operate in both the clockwise and counterclockwise directions of rotation. Direction of rotation shall be determined while facing the drive shaft of the pump or motor.

3.5.5 Vane positioning and tip sealing (class III pumps). When specified (see 6.2), there shall be no damage or cavitation to the class III pumps operating under conditions of high inlet fluid viscosity (860 centistrokes [cSt]).

3.5.6 Internal auxiliary or charge pumps (class I and II closed-loop pumps). Unless otherwise specified (see 6.2), internal auxiliary or charge pumps that provide main pump suction replenishment flow rate and servo-loop control pressure for class I and II pumps in closed-loop applications shall be operated from the main pump drive shaft. Means shall be provided to support filtration of the pump discharge (see 6.2).

3.5.7 Limit of protective coating finish. Application of protective coating finish (see 3.8.1) for external surfaces, as required by 3.2.1, shall avoid surfaces internal to seals and those in contact with hydraulic fluid.

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3.6 Interface, interoperability, and compatibility requirements.

3.6.1 Pressure pulsation or ripple (pumps). Discharge pressure pulsations or ripple effects generated in interfacing hydraulic systems and components by pumps shall be not greater than ±5 percent of the set pressure when operating pumps at full stroke with pressure settings at 25 percent, 50 percent, 75 percent, and 100 percent of designed working pressure.

3.6.2 Noise. Unless otherwise specified (see 6.2), airborne noise and structureborne noise produced by pumps or motors when operating at designed working pressure and flow rate shall be not greater than the following criteria.

3.6.2.1 Airborne noise acceptance criteria. Sound pressure levels shall be in accordance with MIL-STD-1474 for grade D shipboard equipment.

3.6.2.2 Structureborne noise acceptance criteria. One-third octave band structureborne noise levels shall be in accordance with MIL-STD-740-2 for type II equipment.

3.6.3 Mechanical drive interface. As specified (see 6.2), mechanical drive interface shall provide a standard dimensional interface with prime mover for pumps and connected load for motors. Type of connection, either splined or keyed, shall be as specified.

3.6.4 Pipe connections. Unless otherwise specified (see 6.2), pipe connections for hydraulic pumps and motors shall be in accordance with MIL-STD-2193.

3.6.5 Port size.

3.6.5.1 Suction-discharge port size (closed-loop pump or motor). The size of ports and pipe flanges shall be such that velocity of the hydraulic fluid through the connected pipe shall not exceed 20 feet per second at designed working flow rate.

3.6.5.2 Suction-discharge port size (open-loop pump). The size of ports and pipe flanges at the designed working flow rate shall be such that velocity of the hydraulic fluid through the discharge-connected pipe shall not exceed 20 feet per second and through the suction-connected pipe shall not exceed 5 feet per second at the designed working flow rate.

3.6.6 Mounting. Unless otherwise specified (see 6.2), mounting shall be as specified in 3.6.6.1 and 3.6.6.2 below.

3.6.6.1 Pump mounting. Pumps shall be designed to be base- or flange-mounted.

3.6.6.2 Motor mounting. Motors shall be designed for flange- or bracket-mounting to a reduction gear case or to the supporting structure of the driven member of the assembly.

3.6.7 Hydraulic fluids. Unless otherwise specified (see 6.2), pumps and motors shall be resistant to, have no deleterious effects upon, and not be degraded by any of the hydraulic fluids in accordance with MIL-PRF-17672, MIL-PRF-17331, or MIL-DTL-32353, as applicable, while maintaining compliance with all requirements of this specification.

3.6.8 Lifting and handling. Attachments for lifting eyes or handles shall be provided when the weight of the hydraulic pump or motor exceeds 50 pounds.

3.6.9 Variable delivery control (class I and II pumps). Design of all class I and II variable delivery pumps shall provide for reversing control.

3.6.9.1 Control type. The control type specified (see 6.2) shall provide for continuous control of class I and II pump delivery from zero to full stroke in either direction without evidence of sluggishness, erratic stroking, or oscillation in the pump's response as viewed from the pump's delivery indicator.

3.6.9.2 Stroke control effort (class I and II pumps with mechanical input). Unless otherwise specified (see 6.2), the torque required to set pump on stroke in either direction shall not exceed 66 inch-pounds. This requirement only applies if the control type requires a mechanical input interface.

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3.6.10 Variable delivery indicators (class I and II pumps). All class I and II pumps shall have delivery indicators to provide a visual indication of the direction and relative displacement of pump delivery.

3.6.11 Pressure regulation (class I and II pumps). When specified (see 6.2), class I and II pumps shall provide a means to regulate main and, if applicable, auxiliary pump discharge pressures in support of various functions, such as pressure compensation, replenishment, and control/servo-valve operation.

3.6.12 Accessibility and adjustment (class I and II pumps). Pump design shall allow access for adjustment of regulative devices, such as pressure compensators, without the need for major disassembly. Range of adjustment shall be continuous. A positive means for maintaining the setting, such as a lock nut, shall also be provided. In no case shall either task (adjustment or locking) require use of special tools.

3.6.13 Motor displacement. Unless otherwise specified (see 6.2), all motors shall have a fixed displacement.

3.6.14 Motor rotational inertia. When specified (see 6.2), motors shall not exceed the allowable moment of inertia.

3.7 Physical characteristics.

3.7.1 Dimensions. When specified (see 6.2), the external envelope dimensions of pumps and motors shall conform to the installation drawing or interface control drawing (ICD). The external configuration of the pump or motor shall conform to the dimensions pertinent to its installation, center of gravity, and interfacing appendages, such as piping and electrical connection points.

3.7.2 Weight. When specified (see 6.2), the dry and wet weights of the completely assembled motor or pump shall conform to the installation drawing or ICD.

3.7.3 Lubrication. Pumps and motors shall be self-lubricated by means of the fluid circulating within the case. Pumps and motors shall operate smoothly and without evidence of damage or a rise in temperature of the case or fluid in excess of 160 °F. Provisions shall be made to permit filling or venting of internal volumes prior to operation or during controlled start-up operation.

3.7.4 Cleanliness. Unless otherwise specified (see 6.2), pumps and motors shall comply with the following cleanliness requirements:

a. Particulate cleanliness shall be ISO 4406 contamination level code 19/17/14 or better.

b. Water content shall not exceed 0.05 percent (500 parts per million) by volume for hydraulic fluids that are not water-based.

3.8 Process requirements.

3.8.1 Finish. The protective coating finish specified in MIL-DTL-24441, MIL-DTL-24441/20 (formula 150), or MIL-DTL-24441/21 (formula 151) is recommended, but not mandatory.

3.8.2 Cleaning. The assembled pump or motor shall be cleaned by running the pump or motor using MIL-PRF-6083, or alternate, hydraulic fluid (see 3.6.7) and appropriate filtration (see 4.6.1) until the required cleanliness level is achieved (see 3.7.4). Results shall be documented.

3.9 Reliability. Unless otherwise specified (see 6.2), pump or motor drive shaft bearings shall provide a minimum ISO 281 L10 rating of 10,000 hours service life at the manufacturer's rated operating conditions of speed and pressure corresponding to the required operating conditions (see 3.4.1).

3.10 Maintainability.

3.10.1 Repair time. The Median Time to Repair (MedTTR) for all shipboard repair actions for a pump or motor shall be not greater than 1 hour. The Maximum Time to Repair (MaxTTR) for each of the shipboard repair actions listed in 4.6.11 for a pump or motor shall be not greater than 3 hours.

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3.10.2 Maintainability features. The pump or motor shall include the following maintainability features:

a. All wear surfaces shall be replaceable or repairable.

b. Component parts that are not functionally interchangeable shall not be physically interchangeable.

c. Line replacement and disassembly of the pump or motor shall not require special tools.

d. Disassembly of the pump or motor shall not require special tools.

3.11 Identification plates. Pumps and motors shall have identification plates that are permanently attached and located for line-of-sight visual access. Identification plates shall be made of brass, nickel-copper alloy, or corrosion-resistant steel in accordance with MIL-DTL-15024, with the following minimum items of information:

a. Product name (e.g., “Pump, hydraulic, axial-piston”).

b. Manufacturer’s name and Commercial and Government Entity (CAGE) Code.

c. Product serial number or date of manufacture.

d. Product drawing or part number.

e. Rated output torque (for motor) or delivery (for pump).

f. Rated working pressure (PSIG).

g. Rated speed (rpm).

h. Fluid.

3.12 Logistics. Logistics data, such as repair parts lists, drawings, and technical manuals, shall be as specified (see 6.2).

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First article inspection. First article inspection shall be performed on one complete pump or motor assembly when a first article sample is required (see 3.1). This inspection shall be accomplished by the examination, analysis, demonstrations, and tests specified in table I.

TABLE I. First article inspection.

Verification	Component	Requirement(s)	Verification method
Examination	Pumps & motors	3.2, 3.5.6, 3.5.7, 3.6.3, 3.6.4, 3.6.5, 3.6.6, 3.6.8, 3.6.10, 3.6.12, 3.7.1, 3.7.2, 3.8.1, 3.11, 3.12	4.4
Environmental conditions			
Salt spray or fog	Pumps & motors	3.3.1	4.6.7.1
Vibration	Pumps & motors	3.3.2	4.6.7.2
Shock	Pumps & motors	3.3.3	4.6.7.3

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TABLE I. First article inspection – Continued.

Performance characteristics			
Operating conditions	Pumps & motors	3.4.1	4.6.3.1
	Motors	3.4.1	4.6.3.2, 4.6.3.3
Pumping characteristics	Pumps	3.4.2.1	4.6.3.1
Motoring characteristics	Motors	3.4.2.2	4.6.3.2
Efficiency			
Mechanical efficiency	Pumps	3.4.3.1	4.6.3.1
	Motors	3.4.3.1	4.6.3.2
Volumetric efficiency	Pumps	3.4.3.2	4.6.3.1
	Motors	3.4.3.2	4.6.3.2
Design/construction			
Endurance	Pumps	3.5.1	4.6.4.1
	Motors	3.5.1	4.6.4.2
Proof pressure	Pumps & motors	3.5.2	4.6.2
Joint tightness	Pumps	3.5.3	4.6.6.1
	Motors	3.5.3	4.6.6.2
Direction of rotation	Pumps	3.5.4	4.6.6.1
	Motors	3.5.4	4.6.6.2
Vane positioning and tip sealing (class III pumps)	Pumps	3.5.5	4.6.5.4
Internal auxiliary charge pumps (class I and II closed-loop pumps)	Pumps	3.5.6	4.6.5.3
Interface, interoperability, and compatibility			
Pressure pulsation or ripple	Pumps	3.6.1	4.6.5.1
Noise	Pumps & motors	3.6.2	4.6.5.2
Hydraulic fluid	Pumps & motors	3.6.7	4.6.1
Variable delivery control (class I and II pumps)	Pumps	3.6.9	4.6.5.5
Accessibility and adjustment	Pumps	3.6.12	4.6.5.5
Rotational inertia	Motors	3.6.14	4.6.5.6
Physical characteristics			
Lubrication	Pumps & motors	3.7.3	4.6.8
Cleanliness	Pumps & motors	3.7.4, 3.8.2	4.6.9
Other requirements			
Reliability	Pumps & motors	3.9	4.6.10
Maintainability	Pumps & motors	3.10	4.6.11

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4.3 Conformance inspection. Conformance inspection of all pumps or motors shall be accomplished by the examination, demonstrations, and tests specified in table II.

TABLE II. Conformance inspection.

Verification	Component	Requirement(s)	Verification method
Examination	Pumps & motors	3.2, 3.5.7, 3.6.3, 3.6.8, 3.6.10, 3.6.12, 3.7.1, 3.7.2, 3.8.1, 3.11, 3.12	4.4
Performance characteristics			
Pumping characteristics	Pumps	3.4.2.1	4.6.6.1
Motoring characteristics	Motors	3.4.2.2	4.6.6.2
Efficiency			
Mechanical efficiency	Pumps	3.4.3.1 1/	4.6.6.1
	Motors	3.4.3.1 1/	4.6.6.2
Volumetric efficiency	Pumps	3.4.3.2 1/	4.6.6.1
	Motors	3.4.3.2 1/	4.6.6.2
Design/construction			
Proof pressure	Pumps & motors	3.5.2	4.6.2
Joint tightness	Pumps	3.5.3	4.6.6.1
	Motors	3.5.3	4.6.6.2
Direction of rotation	Pumps	3.5.4	4.6.6.1
	Motors	3.5.4	4.6.6.2
Internal auxiliary charge pumps (class I and II closed-loop pumps)	Pumps	3.5.6	4.6.5.3
Interface, interoperability, and compatibility			
Hydraulic fluid	Pumps & motors	3.6.7	4.6.1
Variable delivery control (class I and II pumps)	Pumps	3.6.9	4.6.6.1
Accessibility and adjustment	Pumps	3.6.12	4.6.5.5
Physical characteristics			
Cleanliness	Pumps & motors	3.7.4, 3.8.2	4.6.9

NOTE:

1/ Mechanical and volumetric efficiency testing at hot oil conditions are not required for conformance verification of production units.

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4.4 Examination. Each pump or motor shall be examined for compliance with the requirements listed in table I and table II, to include design, materials, visible defects, coatings, dimensions, markings, and workmanship. This element of inspections shall encompass all visual examinations and dimensional requirements. Noncompliance with any specified requirement as revealed by visual inspection shall constitute cause for rejection.

4.5 Test conditions. Unless otherwise specified (see 6.2), test conditions shall conform to 4.5.1, 4.5.2, and 4.5.3.

4.5.1 Test operating speed. Test pump or test motor shaft speed shall be maintained constant, as practical conditions permit. Operating speed shall be rated speed.

4.5.2 Suction pressure. Suction pressure for pumps shall be maintained in the range of 10 inches of mercury (inHg) to 20 PSIG.

4.5.3 Suction temperature. Suction temperature shall be as specified in 4.5.3.1 for testing at normal oil conditions and as specified in 4.5.3.2 for testing that requires hot oil conditions. For closed-loop pumps that include individual suctions or inlets for both the main pump and the auxiliary or charge pump, each suction shall be maintained within the same temperature range specified herein, as applicable for the test being performed.

4.5.3.1 Normal oil conditions. Suction temperature shall be maintained in the temperature range of 95 to 105 °F.

4.5.3.2 Hot oil conditions. Operational testing at hot oil conditions shall be performed at suction oil temperatures in the range of 150 to 160 °F.

4.6 Methods of inspection.

4.6.1 Hydraulic fluid. Unless otherwise specified (see 6.2), test hydraulic fluid shall be ISO viscosity grade 32, 46, or 68 to determine conformance to 3.6.7. The fluid shall be filtered through an SAE J2321/2-7 filter element or an equivalent filter element having a minimum filtration ratio of 75 at 7 microns.

4.6.2 Hydrostatic pressure. Pumps and motors shall be tested to determine compliance with 3.5.2.1 and 3.5.2.2. The pump or motor and fittings shall be subjected to hydrostatic test pressures at the main flow ports (see 3.5.2.1) and the case drain port (see 3.5.2.2) for a minimum of 5 minutes for each port. Case drain ports may be vented to tank during the test of main flow ports to prevent damage of the case drain.

4.6.3 Performance.

4.6.3.1 Pumping and efficiency tests. Pumps shall be tested to determine compliance with 3.4.2.1, 3.4.3, 3.5.3, and 3.5.4. The test pump shall be operated at the applicable conditions listed below (see 4.6.3.1.1) to determine measured output flow rate, input torque, and input speed. The resultant data shall be recorded for each operating case on a discharge pressure base. Derived mechanical and volumetric efficiency shall be determined for the conditions of full pump stroke with 100 percent and 150 percent of designed working pressure.

4.6.3.1.1 Operating conditions for pumping and efficiency tests.

4.6.3.1.1.1 Variable displacement pump conditions. The pump shall be operated at ¼, ½, ¾, and full pump stroke in each direction and discharge pressure regulated at 25 percent, 50 percent, 75 percent, 100 percent, and 150 percent of designed working pressure at the normal oil temperature conditions specified in 4.5.3.1. Sufficient time shall be allowed for measured parameters to stabilize at each increment of pump stroke and discharge pressure to permit data collection as specified in 4.6.3.1. The pump shall be operated at full pump stroke in each direction and discharge pressure regulated at 25 percent, 50 percent, 75 percent, and 100 percent of designed working pressure at hot oil conditions (see 4.5.3.2) with sufficient time for measured parameters to stabilize at each increment of discharge pressure to permit data collection as specified in 4.6.3.1.

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4.6.3.1.1.2 Constant displacement pump conditions. The pump shall be operated at full pump stroke with discharge pressure regulated at 25 percent, 50 percent, 75 percent, 100 percent, and 150 percent of designed working pressure at the normal oil temperature conditions (see 4.5.3.1). The pump shall be operated at full pump stroke and discharge pressure regulated at 25 percent, 50 percent, 75 percent, and 100 percent of designed working pressure at hot oil conditions (see 4.5.3.2). Sufficient time shall be allowed for measured parameters to stabilize at each increment of discharge pressure to permit data collection as specified in 4.6.3.1.

4.6.3.2 Motoring and efficiency tests. Motors shall be tested to determine compliance with 3.4.2.2, 3.4.3, 3.5.3, and 3.5.4. The test motor shall be operated at the conditions listed below (see 4.6.3.2.1) to determine measured input flow rate, output torque, and output speed. The resultant data shall be recorded for each operating case on a suction pressure base. Derived mechanical and volumetric efficiency shall be determined for the conditions of rated speed with 100 percent and 150 percent of designed working pressure.

4.6.3.2.1 Operating conditions for motoring and efficiency tests. The motor shall be operated at rated speed and motor suction pressure regulated at 25 percent, 50 percent, 75 percent, 100 percent, and 150 percent of designed working pressure at the normal oil temperature conditions (see 4.5.3.1). Sufficient time shall be allowed for measured parameters to stabilize at each increment of suction pressure to permit data collection as specified in 4.6.3.2. The motor shall be operated with motor suction pressure regulated at 100 percent of designed working pressure for 10 minutes at hot oil conditions (see 4.5.3.2).

4.6.3.3 Minimum constant speed test (motor). The test motor shall be operated under the test conditions listed in 4.5.1 at the normal oil temperature conditions specified in 4.5.3.1. The motor shall be operated at the applicable speed listed in 3.4.1.2.2 for a period of 10 minutes against a maximum of 20 percent of the rated torque to track the motor speed for smooth or steady rotation. Visible indications of motor cogging shall be observed and oscillographic recordings shall be taken for at least 5 minutes to determine peak percentage change in angular velocity from the applicable speed listed in 3.4.1.2.2.

4.6.4 Design/construction.

4.6.4.1 Endurance test (pump).

4.6.4.1.1 Variable displacement pump endurance test. The test pump shall be operated under the test conditions listed in 4.5.1 and 4.5.2 at the normal oil temperature conditions specified in 4.5.3.1. With the test pump operating at designed working pressure, the pump stroke shall be varied on the following schedule: ¼, ½, ¾, and full stroke. At each increment of pump stroke, the pump shall be cycled from one side of neutral to the other every 15 minutes (four cycles per hour) for 25 hours. Output flow rate, input torque, input speed, and mechanical and volumetric efficiency shall be determined for each 15-minute interval of operation.

4.6.4.1.2 Constant displacement pump endurance test. The test pump shall be operated under the test conditions listed in 4.5.1 and 4.5.2 at the normal oil temperature conditions specified in 4.5.3.1. Operate the test pump at designed working pressure for 100 hours. The pump shall be cycled from one side of neutral to the other every 15 minutes. Output flow rate, input torque, input speed, and mechanical and volumetric efficiency shall be determined for each 15-minute interval of operation.

4.6.4.2 Endurance test (motor). The test motor shall be operated under the test conditions listed in 4.5.1 and 4.5.2 at the normal oil temperature conditions specified in 4.5.3.1. Operate the test motor at rated speed and design working pressure for 100 hours. The motor shall be cycled from one side of neutral to the other every 15 minutes. Input flow rate, output torque, output speed, and mechanical and volumetric efficiency shall be determined for each 15-minute interval of operation.

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4.6.5 Interface, interoperability, and compatibility.

4.6.5.1 Pressure pulsations or ripple (pump). The test pump shall be operated under the test conditions listed in 4.5.1 and 4.5.2 at the normal oil temperature conditions specified in 4.5.3.1. With the test pump operating at full stroke, the discharge pressure shall be set at 25 percent, 50 percent, 75 percent, and 100 percent of designed working pressure for 10 minutes at each setting. At each setting, the maximum deviation of pressure shall be determined from a mean of the oscillographic record of the pulsation pattern. Pulsation instrumentation shall be located in the discharge line at approximately 3 feet downstream of pump outlet and shall provide a frequency response that is at least 5 to 10 times the highest frequency component in the pressure signal.

4.6.5.2 Noise tests. Unless otherwise specified (see 6.2), the pump or motor shall be subjected to the airborne noise test of the Shipboard Equipment Noise Requirements appendix of MIL-STD-1474 and the structureborne noise test of MIL-STD-740-2.

4.6.5.3 Pumping test (internal auxiliary or charge pumps). Internal auxiliary or charge pumps for class I and II pumps in closed-loop applications shall be tested to determine compliance with 3.5.6. The test pump shall be operated under the test conditions listed in 4.5.1 and 4.5.2 at the normal oil temperature conditions specified in 4.5.3.1. Test procedure shall verify pressure and flow performance of the internal auxiliary or charge pump when the main pump is delivering full volume at the designed working pressure.

4.6.5.4 Vane positioning and tip sealing test (class III pumps). The test pump shall be operated under the test conditions listed in 4.5.1 and 4.5.2 with oil temperature conditions regulated to obtain an inlet fluid viscosity of 860 cSt minimum. Unless otherwise specified (see 6.2), with an initial pump inlet cold soak to provide a starting fluid viscosity of approximately 860 cSt, the pump shall be started and accelerated to rated speed in not more than 10 seconds. The pump stroke shall be uniformly increased and the pressure discharged over a period of 10 minutes to 25 percent of rated conditions. The pump delivery shall be recorded and indications of cavitation shall be observed.

4.6.5.5 Variable delivery controls test (class I and II pumps). The test pump shall be operated under the test conditions listed in 4.5.1 and 4.5.2 at the normal oil temperature conditions specified in 4.5.3.1. Class I and II pump control features shall be tested to determine compliance with 3.6.9. Testing shall determine the pump's response characteristics (see 3.6.9.1) and, if applicable, the maximum effort required to manually move the stroke adjusting mechanism from zero to full stroke in either direction when working at 25 percent, 50 percent, 75 percent, and 100 percent of designed working pressure.

4.6.5.6 Rotational inertia (motors). An analysis shall be provided to demonstrate moment of inertia does not exceed specified limit (see 3.6.14).

4.6.6 Dynamic performance.

4.6.6.1 Dynamic performance testing (pumps). Pumps shall be tested to determine compliance with 3.4.2.1, 3.4.3.1, 3.4.3.2, 3.5.3, 3.5.4, and, for class I and II variable delivery pumps, 3.6.9. The test pump shall be operated under the operating speed, suction pressure, and normal oil temperature conditions as specified in 4.5.1, 4.5.2, and 4.5.3.1, respectively, at the designed working pressure (see 6.5.1). Testing shall include operation at rated speed and designed working pressure. At least 5 minutes of operation shall be conducted. For class I and II variable delivery pumps, stroke control shall be cycled full volume above center to full volume below center. Performance shall be observed for any of the following unacceptable results:

a. Insufficient delivery (see 3.4.2.1).
b. Insufficient mechanical efficiency (see 3.4.3.1).
c. Insufficient volumetric efficiency (see 3.4.3.2).
d. Excessive external leakage (see 3.5.3).
e. Incorrect direction of rotation (see 3.5.4).
f. Oscillation, sluggishness, or erratic movement of the pump control (see 3.6.9).

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4.6.6.2 Dynamic performance testing (motors). Motors shall be tested to determine compliance with 3.4.2.2, 3.4.3.1, 3.4.3.2, 3.5.3, and 3.5.4. The test motor shall be operated under the operating speed, suction pressure, and normal oil temperature conditions as specified in 4.5.1, 4.5.2, and 4.5.3.1, respectively, at the designed working pressure (see 6.5.1). Testing shall include operation at rated speed, corresponding inlet flow rate, and designed working pressure. At least 5 minutes of testing shall be conducted. Performance shall be observed for any of the following unacceptable results:

a. Insufficient torque output (see 3.4.2.2).

b. Insufficient mechanical efficiency (see 3.4.3.1).

c. Insufficient volumetric efficiency (see 3.4.3.2).

d. Excessive external leakage (see 3.5.3).

e. Lack of capability to rotate in both the clockwise and counterclockwise directions (see 3.5.4).

4.6.7 Environmental conditions testing.

4.6.7.1 Salt spray or fog. The pump or motor shall be subjected to the salt fog test method specified in MIL-STD-810 for 48 hours. After exposure and cleaning, the pump or motor shall be inspected for signs of degradation. The base metal of the pump or motor shall not be visible through the coating or finish. There shall be no evidence of blistering, softening, separation from the base metal, corrosion products, or other coating failures. When more than one design uses the same process for surface preservation, only one pump or motor shall require this test, provided the same materials are used.

4.6.7.2 Vibration tests.

4.6.7.2.1 Environmental vibration. Pumps and motors shall meet the type I environmental vibration criteria of MIL-STD-167-1 to determine compliance with the applicable requirements of 3.3.2.

4.6.7.2.2 Internally excited vibration. Pumps and motors shall meet the type II balance criteria of MIL-STD-167-1 to determine compliance with the applicable requirements of 3.3.2.

4.6.7.3 Shock test. The pump or motor shall be approved by test or extension in accordance with MIL-DTL-901. When approved by test, the pump or motor shall be subjected to the type A high-impact shock test in accordance with MIL-DTL-901 to the specified shock grade and class (see 6.2). During the test, the unit shall be operated at designed working pressure. The unit shall be mounted on the table of the shock machine using fitted foundation bolts. Mounting shall simulate shipboard conditions insofar as practicable. Post shock test inspection shall include all functional and operational tests to demonstrate that the specified requirements are met.

4.6.8 Lubrication. The pump or motor shall be operated for 30 minutes continuously at the designed speed specified in 3.4.1.2.1 and normal oil temperature conditions specified in 4.5.3.1. Class I and II variable delivery pumps shall be tested at the zero-stroke position, where no oil shall be pumped and the head of oil of the suction port shall not exceed 5 feet. At 5-minute intervals during this test, the pump or motor shall be brought to a full stop and immediately restarted. Pump or motor case surface and fluid temperature rise shall be recorded and verified to be in compliance with 3.7.3.

4.6.9 Cleanliness test. The pump or motor fluid sample shall be analyzed in accordance with ASTM D6304 and with an automatic particle counter calibrated to ISO 11171 to determine compliance with 3.7.4 and 3.8.2.

4.6.10 Reliability verification. Analysis shall be conducted to demonstrate satisfactory reliability (see 3.9).

4.6.11 Maintainability demonstration. Maintainability demonstrations shall be performed to verify compliance with required MedTTR and MaxTTR (see 3.10). The demonstrations shall include replacing pump or motor, replacing pump or motor drive shaft seal, and replacing pump or motor drive shaft bearing. Only the corrective maintenance procedures described in technical manuals or product literature shall be used.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be in accordance with the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Hydraulic pumps and motors in accordance with this specification are intended for use with hydraulic systems onboard Naval ships. Hydraulic pumps and motors procured to this specification are form, fit, and function replacements for prior revisions of this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Class required (see 1.2).

c. When first article inspection is required (see 3.1).

d. Shock grade and class (see 3.3.3 and 4.6.7.3).

e. Operating conditions if other than as specified (see 3.4.1).

f. Designed working or rated speed required (see 3.4.1.2.1).

g. Pumping and motoring characteristics required, to include flow-rate output and torque output (see 3.4.2.1 and 3.4.2.2).

h. Mechanical and volumetric efficiency, if other than specified (see 3.4.3).

i. Proof pressure if other than specified (see 3.5.2).

j. Direction of pump rotation, if other than specified (see 3.5.4).

k. When a vane positioning and tip-sealing requirement or alternative test procedure is required (see 3.5.5 and 4.6.5.4; for guidance, see 6.4.5).

l. Discharge flow rate or differential pressure for internal auxiliary or charge pumps if other than specified (see 3.5.6).

m. Internal auxiliary or charge pump discharge filtration requirements (see 3.5.6).

n. Noise requirements or noise test procedure if other than specified (see 3.6.2 and 4.6.5.2).

o. Mechanical drive interface dimension and type of connection required (see 3.6.3).

p. Pipe connections, if other than specified (see 3.6.4).

q. Mounting method required if other than specified (see 3.6.6).

r. Hydraulic fluid if other than specified (see 3.6.7, 4.6.1, and 6.4.3).

s. Variable delivery control type required (see 3.6.9.1 and 6.4.4).

t. Stroke control effort for class I and II variable delivery pumps with mechanical input, if other than specified (see 3.6.9.2).

u. When pressure regulation is required (see 3.6.11).

v. Motor displacement, if other than fixed (see 3.6.13).

w. Motor rotational inertia, when specified (see 3.6.14).

x. Dimensions and weight required (see 3.7.1 and 3.7.2).

y. Fluid cleanliness requirements, if other than specified (see 3.7.4).

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z. When another service life for bearings is required (see 3.9 and 6.4.7).

aa. When logistics data, such as a repair parts list, drawing, or technical manual is required (see 3.12 and 6.4.7).

bb. Test conditions if other than as specified (see 4.5).

cc. Packaging requirements (see 5.1, 6.3.1, 6.3.2, and 6.3.3).

6.3 Preservation, packing, and marking.

6.3.1 Preservation. Following pre-shipment cleaning processes (see 3.8.2), the pump or motor should be filled with a preservative fluid, such as MIL-PRF-6083, to coat interior surfaces, drained, and all ports capped to prevent introduction of contaminants during packaging or shipment (see MIL-P-16789). Care should be taken to prevent entry of contamination into the pump or motor during preservation and handling processes.

6.3.2 Packing. Pumps or motors should be packaged in accordance with ASTM D3951 and MIL-P-16789. The choice of the container should be consistent with the weight and volume of the pump or motor. External surfaces and component projected surfaces, such as ports or indicators, should be cushioned and securely packaged within the shipping container.

6.3.3 Marking. Marking of packing of interior packages and shipping containers for pumps or motors should be in compliance with MIL-STD-129. Tags or shipping container labeling should contain instructions to flush and drain residual preservative fluid prior to installation of the pump or motor. The date of preservation should be indicated on tags or shipping labels with instructions that internal preservation afforded by preservative fluid is limited to 1 year.

6.4 Guidance information.

6.4.1 Corrosion protection guidance for metals. For guidance in the corrosion protection of metals, refer to the discussions of plated coatings, cadmium and zinc plating, aluminum alloys, and ferrous alloys contained in MIL-STD-2193.

6.4.2 Guidance on factor of safety. Stresses under full-load conditions should not exceed 35 percent of the minimum yield point of the material used and, under extreme loading (for example, impact loading), the combined stresses should not exceed 70 percent of the minimum tensile yield point (typ) of the material used. The following yield points may be assumed:

a. Direct shear – 60 percent typ

b. Torsional shear – 60 percent typ

c. Static bearing – 160 percent typ

6.4.3 Guidance on alternate test fluid selection. Selection of the test fluid should take into consideration the following factors: cost of replacing the existing test facility test fluid, application fluid specification, and component fluid viscosity recommendations of NFPA/T2.13.13.

6.4.4 Information on variable delivery control types. Market research of variable delivery class I and II pumps indicates that current technology provides a wide variety of means to control the position of the pump's swash plate actuator and the stroke of the pump. A representative selection from various Navy applications of class I and II pumps is provided below.

6.4.4.1 Mechanical input to rotary valve. This type of control consists of external mechanical input to a rotary valve that ports control oil flow to the swash plate actuator for control of pump stroke. Varying the mechanical input angle varies pilot valve rotary position in either clockwise or counterclockwise direction and adjusts both direction and volume of pump discharge flow rate. This action also opens a slide valve to port servo oil to and from the swash plate actuator that positions the swash plate rocker cam or yoke. When the cam reaches the position equal to that of the mechanical input to the rotary pilot valve, this valve closes to lock the cam position. The pilot valve is spring-biased to blocked center position to terminate swash plate movement when command input returns to neutral position.

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6.4.4.2 Hydraulic pilot pressure input to linear valve. This type of control consists of external hydraulic pilot pressure input to a stroke control valve, which ports control oil to the swash plate actuator for control of pump stroke. Varying the pilot pressure against spring bias on the stroke control valve spool produces proportional movement of the swash plate actuator. As the swash plate actuator strokes, this motion causes a feedback spool valve tracking this motion to continuously meter control oil supply away from the stroke control spool valve until the required pump stroke position is reached and swash plate movement is stopped.

6.4.4.3 Discharge pressure signal input to compensator control valve. This type of control relies upon a combination of directional control valves, relief valves, solenoid valves, and orifices to sense the pump discharge pressure and make adjustments in pump stroke to conserve the pump's power consumption during pump start up and as the load builds up downstream pressure.

6.4.4.4 Hydraulic pilot pressure signal input to servo valve. This type of control uses a hydraulic input signal to operate a spring-centered four-way servo valve, which uses mechanical feedback from the swash plate actuator for closed-loop control. As input hydraulic pilot pressure shifts the servo valve, it ports control oil to either side of the swash plate actuator, which moves the swash plate through an angular rotation that corresponds to the direction and volume of the pump's displacement. Mechanical feedback from the swash plate actuator to the four-way servo valve causes the servo valve to cut back on flow to the swash plate actuator until the angular position of the pump swashplate matches the input hydraulic pilot pressure signal.

6.4.4.5 Electrical signal input to servo valve. This type of control uses a small current electrical signal input from an external closed-loop processor to an electro-hydraulic servo valve torque motor to shift the servo valve spool and port control oil flow to and from the swash plate actuator. The direction and amount of commanded pump stroke is determined by the polarity and amount (amperage) of the input current signal. A feedback potentiometer is used to monitor swash plate or pump stroke position and report this stroke angle information as a small voltage feedback signal to an external closed-loop processor. The polarity (±) of the potentiometer's output signal indicates the direction of pump discharge and the potential (voltage) indicates the pump stroke. For a steering control system, this potentiometer signal as well as the rudder angle command and feedback signals would be used by an external closed-loop processor to generate servo valve command signals for pump delivery.

6.4.4.6 Torque or horsepower limiter control. This type of control compares an input signal from pump outlet port pressure to mechanical feedback of pump stroke to regulate pump output fluid power to a setting that is conservatively within the limits for the desired maximum pump input horsepower or the maximum torque at constant pump speed. Torque limiter control overrides primary pump delivery control, as necessary, to maintain a cap on input torque demand. Torque limiter control is often used in conjunction with compensator pressure limiting control to maximize pump efficiency given a limit on prime mover input power.

6.4.5 Vane positioning and tip sealing guidance. Market research indicates that major vane hydraulic pump suppliers advertise safe pump start-up operations with initial high viscosity (>860 cSt) inlet fluid conditions, provided the circulating fluid is allowed to warm up under controlled light off conditions. In general, deck machinery hydraulic systems use sump heaters to avoid adverse cold weather start-up situations. If so, 3.5.5 and 4.6.5.4 need not be specified for vane pump acquisition. If not, the pump supplier should be consulted regarding demonstration of cold-start pump priming capability.

6.4.6 Service life expenditure. The expenditure of bearing service life can be determined based on actual percentage of time the pump or motor operates at manufacturer's rated operating conditions. Assuming that a pump is running at these conditions for 25 percent of its operating time during a daily 8-hour period for 150 days per year over a continuous 7 years of service, the accumulated service time at manufacturer's rated operating conditions would be 2,100 hours. If the bearing life rating or L10 value of the pump or motor at the manufacturer's rated operating conditions is 10,000 hours, the percent expenditure of bearing life rating based on the above operating conditions is 2,100/10,000 or 21 percent. The L10 value means that 90 percent of the bearings survive, at a minimum, the number of L10 hours calculated for bearing life based on operating speed, system pressure, fluid viscosity, and fluid contamination. Based on the percentage of bearing life expenditure, the specified L10 value can be adjusted up or down.

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6.4.7 Logistics data guidance. The following guidance for acquisition of logistics data is provided.

6.4.7.1 Repair parts list. A listing of onboard spare parts for a pump or motor should include those replacement items, such as seals and bearings, that support shipboard repair actions. In addition, a listing of spare parts for a pump or motor, which are required to support off-hull or depot level repair actions, should be provided.

6.4.7.2 Drawings. Outline, assembly, and, as necessary, subassembly drawings for a pump or motor should be furnished in sufficient detail to support both onboard and depot-level repairs and testing. Performance curves should also be included in the drawing submittal.

6.4.7.3 Technical manuals. Technical manuals should be provided that provide a functional description of the pump or motor, operational aspects of the design, and necessary information for preventative and corrective maintenance.

6.5 Definitions.

6.5.1 Designed working pressure. A term used to denote the nominal pressure rating assigned by the manufacturer of the pump or motor for continuous service and provided in units of PSIG.

6.5.2 Maximum design pressure. A term used to denote the pressure permitted for intermittently applied loads (normally 150 percent of designed working pressure) and provided in units of PSIG.

6.5.3 Motor mechanical efficiency. A term used to denote the ratio of the mechanical output torque measured at the shaft of the motor to the calculated and theoretical output torque of the motor at the conditions tested, provided as a percentage.

6.5.4 Motor volumetric efficiency. A term used to denote the ratio of the measured input flow rate of the motor to the calculated theoretical flow rate of the motor at the conditions tested, provided as a percentage.

6.5.5 Pump mechanical efficiency. A term used to denote the ratio of the calculated theoretical input torque to the pump at the conditions tested to the mechanical input torque measured at the shaft of the pump, provided as a percentage.

6.5.6 Pump volumetric efficiency. A term used to denote the ratio of the measured output flow rate of the pump to the calculated theoretical flow rate of the pump at the conditions tested, provided as a percentage.

6.6 Subject term (key word) listing.

Axial piston

Pressure compensator

Radial piston

Valve, replenishment

Vane

Variable delivery control

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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